YUVRAJ SINGH BHADORIA

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Gurgaon, India | Open to remote

Summary

GPU kernel / LLM inference engineer. I build and benchmark LLM serving stacks: custom Triton & CUDA kernels, quantization, and vLLM. Profiler-guided optimization cut production inference latency **62%** (800ms → <300ms) and cost **25%** for a 10,000+ employee deployment.

Skills

- **GPU & Kernels:** Triton (portable kernels, compiled to CUDA), CUDA, tensor cores, FlashAttention, PagedAttention
- **Profiling & Measurement:** NSight Compute (ncu), NSight Systems, L2 cache hit rate, L2 throughput, occupancy, CUDA-event timing
- **Serving & Inference:** vLLM, SGLang, TensorRT-LLM, llama.cpp, quant (INT4 / INT8, KV-cache)
- **Post-Training:** SFT, LoRA / QLoRA, DPO, GRPO (verifiable rewards)
- **Languages & Tools:** C++ (C++17), Python, Git, Linux, PyTorch

Experience

ML Engineer, LLM Inference *May 2026 – Present*
kipi.ai (WNS), Gurgaon

- Optimize **LLM inference** for enterprise GenAI on Snowflake Cortex (serving-config + prompt-efficiency tuning): cut **latency 3**× (3 s → <1 s) and raised **concurrency 4**× (50 → 200) for a Fortune 500 client, at frozen quality
- Built production **LLM review pipelines** on **Cortex Complete** (mistral-large2): Snowpark procedures scoring 1–5 (≥4 = PASS) against project plan, RAID, and 4-week history for live accounts
- Cut per-review cost & latency via file-level **dedup** (extract once, review N use-cases), 480K-char **prompt token-budgeting**, and per-use-case **fail-fast isolation**; unit-tested end to end

ML Engineer *Jun 2025 – May 2026*
Bank of America, Gurgaon

- Optimized **GPU kernels** for LLM serving (vLLM) of enterprise assistant **Ericca** (10,000+ employees): **NSight Compute** tuning cut **latency 62%** (800 → <300 ms) and API **cost 25%** at frozen quality
- Applied measure → profile → optimize → re-measure to improve **L2 cache hit rate**, occupancy, and memory throughput; documented when profiler metrics guide kernel decisions
- Tuned **PagedAttention** (KV-cache) and continuous batching; implemented **INT4/INT8 quantization** plus model selection across **vLLM / SGLang / TensorRT-LLM** with CUDA-event timing

Senior Data Scientist (Contract) *Feb 2025 – Jun 2025*
Policybazaar, Gurgaon

- Applied **LLM inference optimization** and **RLHF-based post-training** to production document workflows; used **C++17** for performance-critical path tuning

Data Scientist *Dec 2022 – Feb 2025*
Infosys, Gurgaon

- Built production ML solutions end-to-end for enterprise clients (experimentation → deployment), with systematic benchmarking and rigorous quality validation

Projects

Triton Attention Kernels Triton · CUDA · FlashAttention · PagedAttention · NVIDIA T4 [Open-Source]

- Built three portable **Triton kernels** (pure-Triton, compiled to CUDA; validated on T4 sm75) covering the LLM serving path: fused **prefill** (FA2-style online softmax), **split-KV flash-decoding**, and **paged KV-cache** attention
- Verified against PyTorch SDPA (fp16, allclose); fused prefill used **10.2**× **less memory** at 4K (0.44 vs 4.46 GB) and fit where naive attention OOMs at 8K; splitting KV **1**→**32** cut decode latency **1.14**→**0.40 ms** at 1K

Production LLM Serving Platform vLLM · FastAPI · Docker · Kubernetes · Helm [Open-Source]

- Built production-grade LLM serving stack: FastAPI gateway + GPU-facing vLLM inference service with **PagedAttention** and continuous batching
- GPU-aware scaling (gateway on CPU, inference on GPU), circuit breakers, multi-replica routing, Prometheus/Grafana observability; **465 automated tests**

Post-Training Stack (SFT / DPO / GRPO) PyTorch · TRL · LoRA / QLoRA · MLflow [Open-Source]

- Built the full **post-training** pipeline: SFT, **DPO**, and **GRPO** with a programmatic **verifier-as-reward** (exact-match math + sandboxed code); verifier logic CPU-testable & CI-tested; Qwen2.5 **GRPO** on GSM8K via TRL (CUDA)

Open Source & Publications

- Author of public **LLM inference optimization blog series** (github.com/YuvrajSinghBhadoria2/llm-inference-blog) on GPU kernels, serving, and speculative decoding

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering *Jul 2018 – May 2022*
Rustamji Institute of Technology, Gwalior, M.P., India CGPA: **8.40 / 10**